US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12410006
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Weber; Adrian

Locking garbage can device

Abstract

The present invention relates to a locking garbage can device primarily comprised of a garbage can body with at least one actuator catch, a lid comprised of at least one rod with at least one actuator magnet, and a housing with at least one housing magnet. The lid is comprised of a magnetic locking mechanism that repels the actuator magnet of the rod via like-poled magnetic fields into a catch located in the garbage can body such that the lid cannot be removed from the body. By turning the housing, the housing spins such that a magnet with an opposite polarity of the actuator magnet of the rod attracts the rod toward the housing such that it disengages the catch and the lid can be lifted from the body.

Inventors: **Weber; Adrian (West Hempstead, NY)**

Applicant: **Weber; Adrian** (West Hempstead, NY)

Family ID: **86767326**

Appl. No.: **18/067765**

Filed: **December 19, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230192400 A1	Jun. 22, 2023

Related U.S. Application Data

us-provisional-application US 63292503 20211222

Publication Classification

Int. Cl.: **B65F1/16** (20060101); **E05B47/00** (20060101)

U.S. Cl.:

CPC **B65F1/1615** (20130101); **E05B47/0038** (20130101);

Field of Classification Search

CPC: B65F (1/1615); E05B (47/0038); E05B (47/004); E05C (9/04)

USPC: 220/230; 220/323; 292/251.5; 292/DIG.11

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2481429	12/1948	Kerr	292/183	B60J 9/02
3596958	12/1970	Bowerman	292/201	E05B 47/0038
4384656	12/1982	McQuiston	220/908	B65F 1/1615
4489851	12/1983	Russ	220/323	B65D 45/28
4534488	12/1984	Boggs	220/323	B65D 45/28
5367891	12/1993	Furuyama	63/3	A44C 5/2057
6622883	12/2002	Wu	220/323	H01L 21/67373
7267378	12/2006	Drumm	292/DIG.60	E05C 19/163
10894643	12/2020	Leary	N/A	A47G 19/2227
11345540	12/2021	DiBartolo	N/A	B65F 1/1623
2007/0228043	12/2006	Perruccio	220/323	B65F 1/1615

Primary Examiner: Smalley; James N

Attorney, Agent or Firm: Brennan, Manna & Diamond, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/292,503, which was filed on Dec. 22, 2021, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to the field of garbage cans. More specifically, the present invention relates to a locking garbage can device primarily comprised of a garbage can body with at least one actuator catch, a lid comprised of at least one rod with at least one actuator magnet, and a housing with at least one housing magnet. The lid is comprised of a magnetic locking mechanism that repels the actuator magnet of the rod via like-poled magnetic fields into a catch located in the garbage can body, such that the lid cannot be removed from the body. By turning the housing, the housing spins such that a magnet with an opposite polarity of the actuator magnet of the rod attracts the rod towards the housing such that it disengages the catch, and the lid can be lifted from the body. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods of manufacture.

BACKGROUND

(2) Outdoor garbage cans can be difficult to secure. Wild animals like rodents, raccoons, opossums, etc., can make their way into a garbage can by opening the lid of the can and ultimately, spread trash around a property. In addition, rodents can carry serious diseases that can easily spread if a rodent infestation in a garbage can is left unattended. As a solution, individuals may try to tape garbage can lids shut or secure the lid via bungee cords or other fasteners. However, this process is cumbersome and impractical because it must be undone and redone every time garbage must be thrown in the garbage can. In addition, temporary fastening solutions may make a garbage can appear unsightly.

(3) Therefore, there exists a long-felt need in the art for an improved garbage can. There also exists a long-felt need in the art for a locking garbage can device with a locking lid that prevents animals from entering the garbage can via dislodging the lid. Further, there exists a long-felt need in the art for a locking garbage can device with a locking lid that can be locked and unlocked quickly and easily. Further, there exists a long-felt need in the art for a locking garbage can device with a locking lid that preserves the appearance of a traditional garbage can.

(4) The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a locking garbage can device. The device is primarily comprised of a garbage can body with at least one actuator catch, and a lid comprised of at least one rod with at least one actuator magnet and a housing with at least one housing magnet. The lid is comprised of a magnetic locking mechanism. When turned in one position, the housing repels the actuator magnet of the rod via like-poled magnetic fields into a catch located in the garbage can body such that the lid cannot be removed from the body. By turning the housing again, the housing spins, such that a magnet with an opposite polarity of the actuator magnet of the rod attracts the rod toward the housing such that it disengages the catch and the lid can be lifted from the body.

(5) In this manner, the locking garbage can device of the present invention accomplishes all of the foregoing objectives and provides a locking garbage can device with a locking lid that prevents animals from entering the garbage can via dislodging the lid. Further, the device can be locked and unlocked quickly and easily. In addition, the locking mechanism is concealed underneath the lid such that the device preserves the appearance of a traditional garbage can.

SUMMARY

(6) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

(7) The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a locking garbage can device. The device is primarily comprised of a garbage can body with at least one actuator catch, and a lid comprised of at least one rod with at least one actuator magnet, and a housing with at least one housing magnet. The body may be any shape known in the art such as, but not limited to, square, cylindrical, etc., wherein at least one inner wall of the body is comprised of at least one actuator catch. In the preferred embodiment, the inner wall has two catches positioned opposite of one another on the inner wall. The lid is preferably the same shape as the opening of the garbage can body and fits over the opening. The lid may further have at least one handle that aids a user in removing the lid from the opening.

(8) The underside of the lid is further comprised of a frame. At least one housing attaches to the frame via at least one axle such that the housing can rotate around the axle by grasping and turning at least one handle on the housing. The housing protrudes outward from the lid via a top opening such that the handle is accessible while the lid is fully seated around the opening. The housing is further comprised of at least one like-poled housing magnet and at least one unlike-poled housing magnet positioned at differing locations on the housing.

(9) The frame is also comprised of at least two guides, wherein each of the two guides is positioned

on either side of the housing. Each guide is preferably semi-circular in shape and secures at least one rod to the frame. In the preferred embodiment, the device has two rods, wherein one of each rod is on each side of the housing and each rod is comprised of at least one actuator magnet at the first end of each rod. The second end of each rod engages the actuator catch of the body.
(10) The device is also comprised of a method of use. First, the lid is placed over the opening of the garbage can body. In this "unlocked" position, the unlike-poled housing magnet is magnetically connected to the actuator magnet of the rod. The unlike-poled housing magnet and the actuator magnet have opposite magnetic fields such that they are attracted to one another. The like-poled housing magnet and the actuator magnet have the same magnetic field such that they repel one another. By turning the housing using the handle, the housing rotates about the axle such that the unlike-poled magnet disengages the actuator magnet, and the like-poled housing magnet aligns with the actuator magnet in opposition. Because the like-poled magnet and actuator magnet repel one another, the rod is propelled into the actuator catch such that the lid cannot be opened, and the device is placed in a "locked" position. When a user desires to open the lid, they can turn the housing using the handle such that the housing rotates about the axle wherein the unlike-poled magnet aligns and re-engages with the actuator magnet and the like-poled housing magnet no longer aligns with the actuator magnet. As a result of the magnetic field formed between the unlike magnet and the actuator magnet, the rod will be propelled back toward the housing such that the rod disengages the lid and the lid can be separated from the body.
(11) Accordingly, the locking garbage can device of the present invention is particularly advantageous as it provides a locking garbage can device with a locking lid that prevents animals from entering the garbage can by dislodging the lid. Further, the device can be locked and unlocked quickly and easily. In addition, the locking mechanism is concealed underneath the lid such that the device preserves the appearance of a traditional garbage can. In this manner, the locking garbage can device overcomes the limitations of existing garbage cans and garbage can securing methods known in the art.
(12) To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:
(2) FIG. **1** illustrates a side cross-sectional view of one potential embodiment of a locking garbage can device of the present invention in a locked position in accordance with the disclosed architecture;
(3) FIG. **2** illustrates a top view of a lid of one potential embodiment of a locking garbage can device of the present invention in accordance with the disclosed architecture;
(4) FIG. **3** illustrates a top view of one potential embodiment of a locking mechanism of a locking garbage can device of the present invention in an unlocked position in accordance with the disclosed architecture; and
(5) FIG. **4** illustrates a flow chart of a method of using one potential embodiment of a locking garbage can device of the present invention in accordance with the disclosed architecture.
DETAILED DESCRIPTION

(6) The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

(7) As noted above, there is a long-felt need in the art for an improved garbage can. There also exists a long-felt need in the art for a locking garbage can device with a locking lid that prevents animals from entering the garbage can via dislodging the lid. Further, there exists a long-felt need in the art for a locking garbage can device with a locking lid that can be locked and unlocked quickly and easily. Further, there exists a long-felt need in the art for a locking garbage can device with a locking lid that preserves the appearance of a traditional garbage can.

(8) The present invention, in one exemplary embodiment, is comprised of a locking garbage can device primarily comprised of a garbage can body with at least one actuator catch, and a lid comprised of at least one rod with at least one actuator magnet, and a housing with at least one housing magnet. The body may be any shape known in the art such as, but not limited to square, cylindrical, etc., wherein at least one inner wall of the body is comprised of at least one actuator catch. In the preferred embodiment, the inner wall has two catches positioned opposite of one another on the inner wall. The lid is preferably the same shape as the opening of the garbage can body such that it fits over the opening. The lid may further have at least one handle that aids a user in removing the lid from the opening.

(9) The underside of the lid is further comprised of a frame, wherein at least one housing attaches to the frame via at least one axle such that the housing can rotate around the axle by grasping and turning at least one handle on the housing. The housing protrudes outward from the lid via a top opening such that the handle is accessible while the lid is fully seated around the opening. The housing is further comprised of at least one like-poled housing magnet and at least one unlike-poled housing magnet positioned at differing locations on the housing.

(10) The frame is also comprised of at least two guides, wherein each of the two guides is positioned on either side of the housing. Each guide is preferably semi-circular in shape. Each guide secures at least one rod to the frame. In the preferred embodiment, the device has two rods, wherein one of each rod is on each side of the housing and each rod is comprised of at least one actuator magnet at the first end of each rod. The second end of each rod engages the actuator catch of the body.

(11) The device is also comprised of a method of use, wherein the lid is first placed over the opening of the garbage can body. In this “unlocked” position, the unlike-poled housing magnet is magnetically connected to the actuator magnet of the rod. The unlike-poled housing magnet and the actuator magnet have opposite magnetic fields such that they are attracted to one another. The like-poled housing magnet and the actuator magnet have the same magnetic field such that they repel one another. By turning the housing using the handle, the housing rotates about the axle such that the unlike-poled magnet disengages the actuator magnet, and the like-poled housing magnet aligns with the actuator magnet in opposition. Because the like-poled magnet and actuator magnet repel one another, the rod is propelled into the actuator catch such that the lid cannot be opened, and the device is placed in a “locked” position. When a user desires to open the lid, they can turn the housing using the handle such that the housing rotates about the axle wherein the unlike-poled magnet aligns and re-engages with the actuator magnet and the like-poled housing magnet no

longer aligns with the actuator magnet. As a result of the magnetic field formed between the unlike magnet and the actuator magnet, the rod will be propelled back toward the housing such that the rod disengages the lid and the lid can be separated from the body.

(12) Accordingly, the locking garbage can device of the present invention is particularly advantageous as it provides a locking garbage can device with a locking lid that prevents animals from entering the garbage can by dislodging the lid. Further, the device can be locked and unlocked quickly and easily. In addition, the locking mechanism is concealed underneath the lid such that the device preserves the appearance of a traditional garbage can. In this manner, the locking garbage can device overcomes the limitations of existing garbage cans and garbage can securing methods known in the art.

(13) Referring initially to the drawings, FIG. **1** illustrates a side cross-sectional view of one potential embodiment of a locking garbage can device **100** of the present invention in a locked position in accordance with the disclosed architecture. The device **100** is primarily comprised of a garbage can body **200** with at least one actuator catch **212**, and a lid **300** comprised of at least one rod **340** with at least one actuator magnet **342** and a housing **350** with at least one housing magnet **356**. In the preferred embodiment of the device **100**, all components of the device **100** are made from a rigid plastic material such as, but not limited to, acrylic, polycarbonate, polyethylene, thermoplastic, acrylonitrile butadiene styrene, low-density polyethylene, medium-density polyethylene, high-density polyethylene, polyethylene terephthalate, polyvinyl chloride, polystyrene, polylactic acid, acetal, nylon, fiberglass, recycled plastic, biodegradable plastic, etc. The plastic material is preferably waterproof and UV-resistant to protect the device **100** from the weather. In a different embodiment, the device **100** may be made from a durable metal such as, but not limited to, stainless steel or aluminum.

(14) The device **100** is comprised of a garbage can body **200**. The body **200** may be any shape known in the art such as, but not limited to, square, cylindrical, etc. The body **200** may be any garbage can shape, size, and/or capacity known in the art. At least one inner wall **210** of the body **200** is comprised of at least one actuator catch **212**, wherein the purpose of the catch **212** will be explained more fully below. In the preferred embodiment, the inner wall **210** has two catches **212** positioned opposite of one another on the inner wall **210**.

(15) FIG. **2** illustrates a top view of a lid **300** of one potential embodiment of a locking garbage can device **100** of the present invention in accordance with the disclosed architecture. The lid **300** is preferably the same shape as the opening **214** of the garbage can body **200**. In the preferred embodiment, the opening **214** and lid **300** are both circular in shape wherein the lid **300** is greater in diameter than the opening **214** such that the lid **300** fits over the opening **300**. The lid **300** may further have at least one handle **310** that aids a user in removing the lid from the opening **214**.

(16) The underside **304** of the lid **300** is further comprised of a frame **320**. The frame **320** may be fixedly or removably attached to the underside **304**. At least one housing **350** attaches to the frame **320** via at least one axle **354** such that the housing **350** can rotate 3600 around the axle **354** by grasping and turning at least one handle **352** on the housing **350**. The housing **350** protrudes outward from the lid **300** via a top opening **302** such that the handle **352** is accessible while the lid **300** is fully seated around the opening **214**. The opening **302** may be comprised of a waterproof gasket **306** that prevents water from entering the body **200** through the opening **302**. The housing **350** is further comprised of at least one like-poled housing magnet **356** and at least one unlike-poled housing magnet **358** positioned at differing locations on the housing **350**.

(17) FIG. **3** illustrates a top view of one potential embodiment of a locking mechanism of a locking garbage can device **100** of the present invention in an unlocked position in accordance with the disclosed architecture. The frame **320** is also comprised of at least two guides **322**, wherein each of the two guides **322** are positioned on either side of the housing **350**. The guide **322** is preferably semi-circular in shape and secures at least one rod **340** to the frame **320**. In the preferred embodiment, the device **100** has two rods **340**, wherein one of each rod **340** is on each side of the

housing **350**. Each rod **340** is comprised of at least one actuator magnet **346** at the first end **342** of each rod **340**. The second end **344** of each rod **340** engages the actuator catch **212** of the body **200**, wherein the catch **212** may be generally L-shaped, circular, or any shape known in the art that receives the second end **344** of the rod **340**.

(18) FIG. **4** illustrates a flow chart of a method of using one potential embodiment of a locking garbage can device **100** of the present invention in accordance with the disclosed architecture. The device **100** is also comprised of a method **400** of use. First, the lid **300** is placed over the opening **214** [Step **402**]. In the “unlocked” position, the unlike-poled housing magnet **358** is magnetically connected to the actuator magnet **346** of the rod **340**. The unlike-poled housing magnet **358** and the actuator magnet **346** have opposite magnetic fields such that they are attracted to one another (i.e., the unlike magnet **358** has a positive field and the actuator magnet **346** has a negative field). The like-poled housing magnet **356** and the actuator magnet **346** have the same magnetic field such that they repel one another (i.e., the like magnet **356** has a negative field and the actuator magnet **346** has a negative field). Then, by turning the housing **350** using the handle **352**, the housing **350** rotates about the axle **354** such that the unlike-poled magnet **358** disengages the actuator magnet **346** and the like-poled housing magnet **356** aligns with the actuator magnet **346** in opposition [Step **404**]. Because the magnet **356** and actuator magnet **346** repel one another, the rod **340** is propelled into the actuator catch **212**. When inside the catch **212**, the rod **340** prevents the lid **300** from being opened (by striking the catch **212** if the lid **300** is pulled upward) and places the device **100** in a “locked” position. When a user desires to open the lid **300**, they can turn the housing **350** using the handle **352** such that the housing **350** rotates about the axle **354**, such that the unlike-poled magnet **358** aligns and re-engages with the actuator magnet **346** and the like-poled housing magnet **356** no longer aligns with the actuator magnet **346**. As a result of the magnetic field formed between the unlike magnet **358** and the actuator magnet **346**, the rod **340** will be propelled back toward the housing **350** such that the rod **340** disengages the lid **300** and the lid **300** can be separated from the body **200** [Step **406**].

(19) In differing embodiments of the device **100**, the magnets **346**,**356**,**358** may be any magnet type known in the art such as, but not limited to, neodymium iron boron magnets, samarium cobalt magnets, alnico magnets, ceramic magnets, ferrite magnets, etc. In the preferred embodiment, the device has two like-poled magnets **356** opposite of one another on the housing **350** and two unlike-poled knob magnets **358** opposite of one another on the housing **350**.

(20) Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein “locking garbage can device” and “device” are interchangeable and refer to the locking garbage can device **100** of the present invention.

(21) Notwithstanding the foregoing, the locking garbage can device **100** of the present invention and its various components can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that they accomplish the above-stated objectives. One of ordinary skill in the art will appreciate that the size, configuration, and material of the locking garbage can device **100** as shown in the FIGS. are for illustrative purposes only, and that many other sizes and shapes of the locking garbage can device **100** are well within the scope of the present disclosure. Although the dimensions of the locking garbage can device **100** are important design parameters for user convenience, the locking garbage can device **100** may be of any size, shape and/or configuration that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

(22) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different

combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.
(23) What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. A locking garbage can device comprising: a body comprised of a first catch and a second catch; a lid having an opening and a handle; a frame removably attached to an underside of the lid; a housing comprised of a first magnet and a second magnet; an axle rotatably attaching the housing to the frame; a first rod comprised of a third magnet positioned on a proximal end of the first rod; and a second rod comprised of a fourth magnet positioned on a proximal end of the second rod; and wherein a distal end of the first rod is configured to engage the first catch and a distal end of the second rod is configured to engage the second catch; and wherein the housing protrudes outward from the lid via the opening and allows the handle to be accessible when the lid is seated around the opening.
2. The locking garbage can device of claim 1, wherein the device is comprised of a waterproof and a UV-resistant plastic.
3. The locking garbage can device of claim 1, wherein the first magnet, the third magnet, and the fourth magnet are comprised of a positive charge.
4. The locking garbage can device of claim 3, wherein the second magnet is comprised of a negative charge.
5. A locking garbage can device comprising: a body comprised of a first catch and a second catch; a lid comprised of an opening, a handle, and a waterproof gasket; a frame removably attached to an underside of the lid; a housing comprised of a first magnet of a first polarity and a second magnet of a second polarity; an axle rotatably attaching the housing to the frame; a first rod comprised of a third magnet positioned on a proximal end of the first rod with a third polarity identical to the first polarity; and a second rod comprised of a fourth magnet positioned on a proximal end of the second rod with a fourth polarity identical to the first polarity; and wherein the first rod is movably secured to the frame via a first pair of guides and the second rod is movably secured to the frame via a second pair of guides; wherein a distal end of the first rod is configured to engage the first catch and a distal end of the second rod is configured to engage the second catch; and wherein the housing protrudes outward from the lid via the opening and allows the handle to be accessible when the lid is seated around the opening.
6. The locking garbage can device of claim 5, wherein the housing spins 360 degrees around the axle.
7. The locking garbage can device of claim 5, wherein the first catch and the second catch are comprised of an L-shape.
8. The locking garbage can device of claim 5, wherein the second polarity opposes the first polarity.
9. The locking garbage can device of claim 5, wherein the first catch and the second catch are positioned opposite one another on an inner wall of the body.

10. The locking garbage can device of claim 5, wherein the body is comprised of a circular opening.
11. The locking garbage can device of claim 5, wherein the lid is circular.
12. The locking garbage can device of claim 10, wherein the lid is larger in diameter than the opening.